

SOLAR-POWERED VEHICLE

Group II

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Introduction

As the demand for environmentally friendly and intelligent transportation grows, the fusion of solar power, 3D printing, and IoT (Internet of Things) technologies is emerging as a game-changer in the development of high-performance vehicles.

The objective is to create a lightweight, energy-efficient, and autonomous or remote controlled vehicle capable of navigating a track efficiently as well as outline the mechanics, materials, and techniques required to develop such a revolutionary vehicle.

Solar racing cars are an integration of renewable solar energy, lightweight materials and smart technologies. The solar-powered car has advanced 3D printing materials, IoT-based sensors and tracking systems. It has environmental benefits such as zero tailpipe emissions which leads to cleaner air and reduced greenhouse gas emissions and has a potential for lower long-term.

Vehicle Mechanics

Solar energy conversion

How solar panels generate electricity:

The photovoltaic effect is the method by which solar panels, sometimes referred to as photovoltaic panels directly convert sunlight into electrical power.

- 1.) **Photovoltaic Cells:** Numerous separate PV cells, usually composed of semiconductor materials like silicon, make up solar panels.
- 2.) **Sunlight Absorption:** Photons, which are light particles, transfer energy to the silicon atoms' electrons when sunlight strikes the photovoltaic cells.
- 3.) **Electron Excitation:** The electrons are excited by this energy and separate from their atoms.
- 4.) **Electric Field Creation:** These free electrons are forced to flow in a single direction by the internal electric field built into the PV cell.
- 5.) **Electric Current:** A direct current of electricity is produced by this electron flux.
- 6.) **Inverter Conversion:** Many households and businesses utilize alternating current power which is produced by passing the DC electricity through an inverter.

Efficiency considerations and best practices:

The ability of a solar panel to convert sunlight into electrical power is known as efficiency.

- **Material Quality:** Efficiency increases with higher-quality silicon and manufacturing techniques.
- **Cell Technology:** In general, monocrystalline panels are more effective than thin-film or polycrystalline ones.
- **Panel Orientation and Tilt:** For optimal exposure panels should be oriented straight toward the sun. The latitude of the site determines the ideal tilt angle.
- **Climatic Conditions:** Efficiency is impacted by temperature panels operate better in colder climates.
- **Shading and Cleaning:** Dirt buildup can impede efficiency and shade limits exposure to sunshine.

Best Practices for Optimal Efficiency:

- **Professional Installation:** Make careful to tilt, align, and install securely.
- **Regular Cleaning:** Clear away debris, dust, and filth to keep performance at its best.
- **Minimize Shading:** Panels should not be positioned in regions with shadow, and trees that provide shade should be pruned.
- **Monitor Performance:** Check the system's output frequently to spot problems early.
- **Proper Ventilation:** To avoid overheating, make sure there is enough airflow surrounding the panels.

(ARENA, 2025)

Battery storage and power management

Battery Storage Options

1. Rechargeable Nickel-Metal Hydride Batteries such as AAA or AA.
2. Works with small solar system.
3. Moderate energy and environmentally friendly.
4. Lithium-Polymer Batteries
5. Higher energy and lightweight.
6. Requires careful charge management to prevent overcharging.
7. Supercapacitors
8. Store and release energy quickly.
9. Useful for intermittent operation.

Power Management System

- Solar Charge Controller such as a charging module as it prevents overcharging and over-discharging and it can regulate power to the motor for smooth operation.
- Diodes for Preventing Reverse Current and Schottky diodes allow current to flow from the solar panel to the battery and prevent backflow.
- Boost or Buck Converters for voltage adjustments with motor a DC-DC boost converter will be used.

Solar Panel to Battery Integration

- Direct Connection as the panel directly powers the motor but works only in sunlight.
- Battery-Supported System as the solar panel charges the battery, which then powers the motor. This provides operation even in low sunlight.

Motors and drive systems:

Best Motor types for solar-powered vehicles

1.Brushless DC Motors:

- These are highly efficient, lightweight and durable which makes them the ideal solar-powered vehicle.
- It uses electronic communications that has full control over speed and torque.
- They require low maintenance, that makes them rely on long-term sustainable operation.

2.High power-to-weight ratio motors:

- Motors with a high power-to-weight ratio are preferred as they maximize energy efficiency while minimizing weight.
- The faster the speed and improved handling makes them critical in environments where their performance is necessary.

3.Integration with solar power systems:

- Motors that can seamlessly integrate with solar power systems are crucial for optimizing energy usage.

The system can convert DC from solar panels to effectively manage motor operation are essential, which makes this energy a game changer.

Power-to-weight ratio considerations:

1.Impact on performance:

- A higher power-to-weight ratio improves acceleration and speed, making it a critical factor for vehicle performance.
- The more weighs the slower the car will especially then it's driving on hills

2.Calculation:

- The ratio is calculated by dividing the motor's (horsepower) power output by the vehicle's weight.
- For example, a motor with 400 kW power and a vehicle weight of 1,800 kg results in a ratio of 4.5 kg/kW.

3.Efficiency:

- Reducing vehicle weight while maintaining motor power enhances fuel efficiency and overall performance.
- They use thinner sheets, more plastic parts, carbon and magnesium

Designing a 3D-printed chassis and components

Aerodynamics and Weight Distribution

1. Aerodynamics:

Aerodynamics focuses on reducing air resistance while ensuring efficient airflow around the chassis, crucial for improving speed, stability, and energy efficiency.

- **Streamlined Shape:**

- Design a smooth, rounded body with minimal sharp edges to reduce turbulence.
- Use teardrop profiles for parts like the roof and sides, as these shapes minimize drag.

- **Air Channels and Vents:**

- Integrate air channels to direct airflow efficiently for cooling electronic components or brakes while simultaneously reducing drag.
- For vehicles requiring thermal management strategically place cooling vents in areas with minimal disturbance to lateral airflow.

- **Lift Management:**

- Incorporate aerodynamic principles such as underbody diffusers or spoilers to counter lift forces that could destabilize small vehicles at even moderate speeds.

2. Weight Distribution:

- Proper weight distribution ensures stability, traction, and performance in acceleration and cornering.

- **Center of Gravity (CoG):**

Aim for a low and central CoG by placing heavier components closer to the center and lower in the chassis this avoids high-mounted weights to prevent rolling or tilting during sharp turns.

- **Balanced Load:**

Ensure even weight distribution between front and rear axles or across all rotors to maintain control and stability.

- **Lightweight Materials:**

Use lightweight yet strong materials in the design such as carbon fiber filaments or composite 3D-printing materials to maintain the ideal weight and performance balance.

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Structural Design Principles for Stability

Load-Bearing and Stress Points:

- Understand where the most significant forces will act on the chassis such as wheel mount points, battery mounts, and joints and reinforce those areas with additional material or structural elements.
- Use simulation tools such as finite element analysis FEA to identify high-stress zones.

Material Selection:

3D-printing materials based on performance requirements:

- **PLA (Polylactic Acid)** is an affordable and easy to use but brittle; not ideal for high-stress applications.
- **Polyethylene Terephthalate Glycol as it is** durable, flexible, and impact-resistant excellent for moderately stressed chassis.
- **Nylon or Carbon-Reinforced Filaments because they have** superior strength, durability and weight savings and it is the best option for high-performance chassis.

Principles of Stability:

◦ **Triangulation:**

Incorporate triangular structures in the design to distribute forces evenly and enhance rigidity. Triangular bracing around corner joints strengthens weak points.

◦ **Torsional Rigidity:**

Ensure the chassis can resist twisting forces caused by uneven terrain or acceleration. Use a box frame or honeycomb design to promote torsional rigidity while keeping the weight low.

◦ **Vibration Damping:**

Include mounts or dampeners within parts of the chassis that house sensitive electronics to minimize vibrations and improve durability.

Modularity:

Design the chassis with removable or replaceable sections for easy maintenance or upgrades, for example, separate components like battery compartments, motor mounts, or suspension could be made modular.

Testing and Iteration:

3D printing allows for rapid prototyping this is an advantage by testing variations of the design. It also performs stress tests and evaluate handling in real-world scenarios to identify areas for improvement, such as weight adjustments or additional reinforcements.

Practical Applications

1. Scale Considerations

- Different design principles for micro, small, and medium-sized vehicles
- Relationship between wall thickness and overall chassis size
- Printing technology limitations and their impact on design decisions

2. Testing Methodologies

- Finite Element Analysis validation before printing
- Physical testing protocols for printed chassis elements
- Iterative design improvement process based on testing results

3. Real-world Implementation Strategies

- Hybrid approaches combining 3D printed elements with traditional materials
- Strategic reinforcement of key areas with metal inserts or carbon fiber overlays
- Design adaptations for specific vehicle types racing vs utility.

IOT sensors for navigation and control

IoT Sensors for Navigation and Control

Types of Sensors

IoT Sensors are hardware components that are used to identify a shift or change in an environment and gather data. These sensors can measure a variety of things such as temperature, motion, pressure and much more, then all this information is shared to all different networks so all devices can interconnect. By collecting data in real-time the sensors allow for predictive maintenance and data-driven insights (Zipit, 2024)

Ultrasonic Sensor

- Ultrasonic sensors produce and discharge ultrasonic waves and change the sound reflected into an electrical signal and establish the distance of the target.
- An ultrasonic sensors most important components are a transmitter that utilizes piezoelectric crystals to create the sound and a receiver to pick up the sound after it has gone to the target and back.
- The primary use of these sensors is to measure distance (Renke, 2024).

Infrared Sensor

- These sensors locate infrared radiation that objects emit then the radiation detected is converted to an electrical signal.
- IR sensors are used to detect infrared radiation, which is released by objects according to their temperature (Cavli Wireless, 2025).

LiDAR Sensor

- A common remote sensing technique called LiDAR uses light in the form of a pulsed laser to identify and classify things.
- These sensors produce accurate, 3-Dimensional (3D) information about the shape of the objects it detects and the characteristics of its surface
- The method of LiDAR creates a three-dimensional (3D) image of the scanned area using laser beams that are within eye-safe range (Petit, 2020)

Obstacle Detection and Avoidance

How sensors work for autonomous navigation

Sensors play a critical role in enabling autonomous navigation systems to operate safely and effectively. These systems are designed to allow machines (such as self-driving cars, drones, or robots) to move through environments without human intervention, and they rely on a combination of sensors to perceive their surroundings, make decisions, and plan their movements.

Types of Sensors Used in Autonomous Navigation

Here are the primary types of sensors that contribute to autonomous navigation systems:

1. LiDAR (Light Detection and Ranging)

How it works: LiDAR sensors emit laser pulses that bounce off objects and return to the sensor. By measuring the time it takes for the laser to return the system calculates the distance to each object building a 3D map of the environment.

Real-life examples:

- Self-driving cars (e.g., Waymo, Tesla): LiDAR helps vehicles detect obstacles, map out roads and identify pedestrians, other vehicles and traffic signals.
- Robots (e.g., robotic vacuum cleaners like Roomba): LiDAR helps these robots navigate around furniture, walls, and other objects within a home or office.

2. Radar (Radio Detection and Ranging)

How it works: Radar sensors use radio waves to detect objects. The sensor emits radio waves, which reflect off objects and return to the sensor. By calculating the time it takes for the wave to return, the system can determine the distance and speed of an object.

Real-life examples:

- Autonomous vehicles (e.g., Ford, BMW): Radar sensors are commonly used in self-driving cars to detect vehicles and obstacles in poor weather conditions, such as rain or fog, where LiDAR and cameras might be less effective.
- Drones: Drones use radar to detect obstacles in their flight path, particularly in conditions with low visibility.

Example algorithms for sensor-based movement.

Autonomous systems rely on algorithms to process sensor data and make movement decisions as algorithms interpret inputs from sensors like cameras, Lidar, radar, IMU, and GPS to ensure safe and efficient navigation. Below are some common algorithms used for sensor-based movement, along with real-world applications

3. Simultaneous Localisation and Mapping (SLAM)

How It Works

SLAM is used for real-time map creation and localisation in unknown environments. It fuses data from Lidar, cameras, IMU, and odometry to build a map while tracking the system's position.

Example Applications

- Waymo's Self-Driving Cars: Uses Lidar-based SLAM to build 3D maps of city streets.
- Boston Dynamics' Spot Robot: Uses SLAM for autonomous movement in buildings.
- DJI Drones: Uses visual SLAM (V-SLAM) for indoor navigation where GPS is unavailable.

4. Deep Reinforcement Learning (AI-Based Navigation)

How It Works

AI-based approaches use reinforcement learning to train autonomous agents to navigate by rewarding good actions and penalizing bad ones.

Example Applications

- Tesla's Full Self-Driving (FSD): Uses AI-trained models to predict and react to traffic.
- Boston Dynamics' Spot Robot: Uses reinforcement learning for better movement strategies in difficult terrain.

5. Fuzzy Logic Controller (Handling Uncertainty in Navigation)

How It Works

Fuzzy logic allows autonomous systems to make decisions in uncertain environments by defining rules based on degrees of truth rather than binary conditions.

Example Applications

- Self-Driving Cars (Lane Keeping Systems): Adjusts steering smoothly based on lane deviation.
- Drones (Autonomous Flight Control): Handles varying wind conditions dynamically.

Autonomous navigation requires multiple algorithms working together to perceive, plan, and move efficiently.

Comparison of Accuracy and Range

	Ultrasonic	Infrared	LiDAR
Accuracy	Up to 1% of measuring range	$\pm 0.5^{\circ}\text{C}$ to $\pm 2^{\circ}\text{C}$ in temperature ,few millimeters to centimeters in distance	0.5 millimetres to 10 millimetres
range	21 meters	Few millimetres	1000 meters and beyond

Motors and Drive Systems

The selection of motors and drive systems plays a vital role in the performance of a solar powered racing vehicle. The motor must efficiently convert stored electrical energy into mechanical movement while maintaining an optimal power-to-weight ratio for speed and endurance.

Best Motor Types for Solar-Powered Vehicles

1. DC Motors – Commonly used due to their simplicity and ease of control.
- 2.Brushless DC Motors – Offer higher efficiency, longer lifespan, and better speed control than brushed DC motors.
3. Servo Motors – Provide precise control but are less common for continuous motion applications.
4. Stepper Motors – Suitable for controlled movements but consume more power, making them less ideal for racing vehicles.

Power-to-Weight Ratio Considerations

- Motor Efficiency – Affects how much of the electrical energy is converted into motion. BLDC motors typically offer efficiency levels of 85–90%.
- Vehicle Weight – Lighter motors reduce overall weight, improving acceleration and energy efficiency.
- Torque and Speed – A balance between torque (force) and speed is necessary to ensure smooth movement on different terrains.
- Gear Ratio Optimization – Gears can be used to adjust speed and torque, optimizing performance for racing conditions.

Drive System Configurations

1. Direct Drive – The motor is directly connected to the wheels, reducing mechanical losses but requiring a powerful motor.
2. Gear Drive – Uses a transmission system to optimize torque and speed, enhancing efficiency.
3. Belt Drive – Provides smooth motion and reduces vibrations but can be less efficient due to slippage

Integration of IoT with Embedded Systems

The combination of the Internet of Things (IoT) and embedded systems has transformed modern technology by enabling real-time data collection, automation, and intelligent decision-making. In the case of a solar-powered 3D-printed smart racing vehicle, IoT enhances efficiency by allowing seamless interaction between sensors and microcontrollers. This integration depends on well-defined communication protocols and effective programming for sensor-based automation.

How Sensors Interact with Microcontrollers

- Role of Sensors in Embedded Systems

Sensors play a crucial role in gathering environmental data, such as speed, temperature, location, and battery status. This data is then transmitted to the microcontroller which processes the information and triggers appropriate actions based on predefined instructions.

- Communication Protocols for Sensor-Microcontroller Interaction

To ensure effective data transmission, wired and wireless communication protocols are used for sensor-to-microcontroller communication.

- Wired Communication Protocols

- I2C (Inter-Integrated Circuit): A two-wire protocol used for connecting multiple low-speed sensors like temperature and humidity sensors.
- SPI (Serial Peripheral Interface): A fast data transfer protocol often used with inertial measurement units (IMUs) and GPS modules.
- UART (Universal Asynchronous Receiver-Transmitter): A serial communication protocol commonly used with GPS and Bluetooth modules.

- Wireless Communication Protocols

- Wi-Fi (ESP8266, ESP32): Enables remote monitoring of vehicle data through IoT connectivity.
- Bluetooth (HC-05, BLE): Facilitates short-range wireless data transfer between the vehicle and mobile devices.
- LoRa (Long Range): Supports long-distance, low-power communication for telemetry applications.

- Data Processing and Automated Responses

Once sensors send data to the microcontroller, it processes the information and takes necessary actions. For example:

- Speed sensors adjust the vehicle's acceleration based on terrain.
- Ultrasonic sensors detect and avoid obstacles.
- GPS modules track the vehicle's location and optimize navigation routes.

Importance of Programming in Sensor-Based Automation

- Embedded Programming for Automation

Programming is key to ensuring smooth sensor-based automation by processing, analyzing, and responding to sensor inputs. Microcontrollers such as Arduino, Raspberry Pi, and ESP32 are programmed using different languages:

- C/C++ (used in Arduino) for precise real-time control.
- Python (used in Raspberry Pi) for data analysis and IoT functions.
- Micro Python for lightweight IoT applications.

- Implementing Sensor-Based Automation

Automation in the smart racing vehicle relies on programming to execute various tasks, such as:

- Real-Time Data Analysis: Processing sensor inputs to adjust vehicle performance dynamically.
- Sensor Fusion: Combining data from different sensors, such as GPS and accelerometers, for accurate navigation.
- Cloud Integration & Data Storage: Using IoT-based programming to upload sensor data to cloud platforms like Firebase, AWS IoT for remote access.
- Predictive Maintenance with Machine Learning: AI-driven analysis of sensor data to predict potential system failures and improve performance.

Efficient communication between sensors and microcontrollers through various wired and wireless protocols enables real-time monitoring and automation. Programming is vital for processing sensor data, allowing the system to make intelligent decisions and automate key functions in the solar-powered 3D-printed smart racing vehicle. By optimizing sensor interaction and utilizing advanced programming techniques, the vehicle can improve performance, conserve energy, and enhance racing capabilities.

Conclusion

Solar racing cars require an integration approach combining renewable energy, advanced manufacturing and IoT technologies. It should have high-efficiency solar panels, be lightweight 3D-printed and have built in smart sensor-based for obstacle detection.

The report covers three main areas:

- The mechanics of solar-powered vehicles and their energy efficiency.
- The role of 3D printing in designing an optimal chassis for racing performance.
- The implementation of IoT sensors for autonomous navigation and obstacle detection.

References

- Green, M. A. (2019). *Solar cells: Operating principles, technology, and system applications*. Prentice Hall.
- Kalghatgi, G. (2021). *Transportation in a net-zero world: Transitioning towards sustainable mobility*. Cambridge University Press.
- Raza, A., Nadarajah, M., & Ekanayake, J. (2020). *Energy storage for sustainable microgrid systems*. Academic Press.
- Wang, Y., & Hu, C. (2022). *Advanced electric vehicle power management and control*. Wiley.
- ARENA, 2025. Australian Renewable Energy Agency. [Online]
- Available at: <https://arena.gov.au/>
- [Accessed 1 April 2025].
- Energy, U. D. o., n.d. Solar Energy. [Online]
- Available at: <https://www.energy.gov/topics/solar-energy>
- [Accessed 1 April 2025].
- SEIA, 2024. Solar Energy Industries Association. [Online]
- Available at: <https://seia.org/>
- [Accessed 1 April 2025].
- Edina.eu (2024) 'Battery Energy Storage System (BESS) | The Ultimate Guide', Edina. Available at: <https://www.edina.eu> (Accessed: 2 April 2025).
- Elum Energy (2024) 'Solar+Storage: Battery types for solar systems', Elum Energy. Available at: <https://www.elum-energy.com> (Accessed: 2 April 2025).
- Basic3DP, E. (2025). How To 3D Print Car Parts Like A Pro? [online] Basic3DP. Available at: <https://basic3dp.com/how-to-3d-print-car-parts-like-a-pro/> Accessed: 2 April 2025
- Walden, D. (2024). Best Material for 3D Printing Car Parts: A Comprehensive Guide - Bell Engineering. [online] Bell Engineering. Available at: <https://bellengineering.net/misc/best-3d-printing-materials-car-parts/> Accessed: 1 April 2025
- Bacchioni, M., Cristillo, F., Patalano, S., & Vitolo, F. (2022). CAE approach for the weight reduction of the chassis of a competition vehicle. *Procedia Structural Integrity*, 37, 95-102. <https://doi.org/10.1016/j.prostr.2022.01.070>
- Chen, X., Liu, Y., & Zhang, J. (2023). Design optimization of lightweight 3D-printed structures with variable-density lattice infill for small electric vehicles. *Composite Structures*, 304, 116363. <https://doi.org/10.1016/j.compstruct.2022.116363>
- Dizon, J. R. C., Espera, A. H., Chen, Q., & Advincula, R. C. (2018). Mechanical characterization of 3D-printed polymers. *Additive Manufacturing*, 20, 44-67. <https://doi.org/10.1016/j.addma.2017.12.002>
- Egan, P. F., Gonella, V. C., Engensperger, M., Ferguson, S. J., & Shea, K. (2017). Computationally designed lattices with tuned properties for tissue engineering using 3D printing. *PloS one*, 12(8), e0182902. <https://doi.org/10.1371/journal.pone.0182902>
- Hållgren, S., Pejryd, L., & Ekengren, J. (2016). 3D data export for additive manufacturing-improving geometric accuracy. *Procedia CIRP*, 50, 518-523. <https://doi.org/10.1016/j.procir.2016.05.046>
- Jiang, J., & Xu, X. (2021). Design optimization of 3D printed plastic parts considering both structural performance and manufacturing cost. *Rapid Prototyping Journal*, 27(1), 149-163. <https://doi.org/10.1108/RPJ-01-2020-0008>
- Xu, L., Wang, L., & Li, H. (2022). Advances in Solar-Powered Vehicles: A Review. *Renewable Energy Research Journal*.
- Smith, J., & Kumar, P. (2021). 3D Printing Technologies in Automotive Engineering. *International Journal of Manufacturing Science*.
- Brown, T. (2020). IoT-Based Vehicle Control Systems. *Journal of Embedded Systems & Automation*.
- Arduino. (2023). Guide to Ultrasonic and LiDAR Sensors in Robotics. Retrieved from www.arduino.cc
- National Renewable Energy Laboratory (NREL). (2023). Solar Panel Efficiency Research. Retrieved from www.nrel.gov

- Jones, A., Patel, R., & White, K. (2021). Additive Manufacturing in Lightweight Automotive Design. *Journal of 3D Printing & Engineering*.
- Williams, R., Chen, D., & Foster, G. (2022). Multi-Material 3D Printing for Structural Applications. *Additive Manufacturing Science Journal*.
- "Internet of Intelligent Things: A Convergence of Embedded Systems and Artificial Intelligence" Authors: Not specified Year: 2024 Source: ScienceDirect URL: <https://www.sciencedirect.com/science/article/pii/S2542660524000945>
- "Research on Embedded-Based IoT Information Data Collection Technology" Authors: Not specified Year: Approximately 2022 Source: ResearchGate URL: https://www.researchgate.net/publication/374739107_Research_on_embedded-based_IoT_information_data_collection_technology
- "Integration of Internet Protocol and Embedded System on IoT Device Automation" Authors: Not specified Year: Approximately 2021 Source: ResearchGate URL: https://www.researchgate.net/publication/355727573_Integration_of_Internet_Protocol_and_Embedded_System_on_IoT_Device_Automation
- "IoT Integration in Embedded Systems" Authors: Dr. Ravva Gurunadha, Dr. Gottapu Appala Naidu, Dr. Mamidipaka Hema, and Mr. China Raju Manda Year: 2024 Source: SAN Publications URL: <https://sanpublications.nobelonline.in/product/iot-integration-in-embedded-systems/>
- "A Survey and Challenges: Embedded System on IoT" Authors: David Williams Year: Approximately 2021 Cavli Wireless, 2025. *Cavli Wireless*. [Online]
- Available at: <https://www.cavliwireless.com/blog/nerdiest-of-things/sensors-in-iot-applications> [Accessed 03 April 2025].
- Petit, F., 2020. *Blickfeld*. [Online] Available at: <https://www.blickfeld.com/blog/what-is-lidar-sensor/> [Accessed 03 April 2025].